

Total No. of Questions : 8]

SEAT No. :

PE-2598

[Total No. of Pages : 2

[6583]-129

T.E. (IT)

MACHINE LEARNING

(2019 Pattern) (Semester - V)(314443)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6 and Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) What do you understand by overfitting data? Gives any two methods to avoid overfitting. [6]
- b) Given the set of values $X = (3, 9, 11, 5, 2)^T$ and $Y = (1, 8, 11, 4, 3)^T$. Evaluate the regression coefficients. [5]
- c) Compare the linear regression model and the logistic regression model. [6]

OR

- Q2)** a) What are the metrics used to validate the result of the regression and explain each? [6]
- b) Distinguish between overfitting and underfitting. How it can affect model generalization. [5]
- c) Discuss linear regression with examples. Explain the role of the hypothesis function in ML. [6]
- Q3)** a) What is Gini impurity? Briefly explain the properties of Gini Impurity. [8]
- b) [CO4, L₁] How does the decision tree handle missing attribute Values? [4]
- c) [CO4, L₂] What do you understand by Information gain? Explain the mathematical formula associated with it. [6]

OR

P.T.O.

- Q4)** a) Write short notes on [12]
i) Conditional probability
ii) Tree Pruning
iii) Bayesian Network
b) What do the root node, leaf node, and parent node represent in a decision tree model? Represent them pictorially. [6]

- Q5)** a) What is the difference between the Manhattan distance and Euclidean distance? [4]
b) Explain in detail Apriori Algorithm. [8]
c) Define the following terms concerning K - Nearest Neighbor Learning [6]
i) Regression
ii) Residual
iii) Kernel Function

OR

- Q6)** a) What is the difference between k-Means and K-Medians and when would you use one over another? [6]
b) What is hierarchical clustering? Explain the issue of connectivity constraints. [8]
c) What are support and confidence measures in data mining? [4]
- Q7)** a) With suitable equations, explain any two types of activation functions used in a neural network. [8]
b) Explain why the initialization process of weight and bias is important for neural networks. [6]
c) Write short notes on Mean square Error. [3]

OR

- Q8)** a) Explain with a neat diagram how neural networks are modeled. [6]
b) Write short notes on [6]
i) Sigmoidal Neuron
ii) McCulloch Pitts Neuron
c) Explain the working of a Perceptron with a neat diagram. [5]

